

SONG PEI-LING

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I am a Master's student in ITBE@TUM, focusing on BIM, semantic modeling, and API development. Experienced in automation, point cloud processing, IoT integration, and real-time visualization to support decision-making across the building lifecycle and advance AEC digitalization.

EDUCATION

- **Technical University of Munich** Munich, Germany
M.Sc. Information Technologies for the Built Environment
– Ministry of Education Government Scholarship to Study Abroad (GSSA) Oct 2024 - Present
- **Chung Yuan Christian University** Taoyuan, Taiwan
B.Arch. Architecture (GPA: 3.77 / 4.0)
– Excellent Performance in Professional Studies Sep 2018 - Jun 2023

PROFESSIONAL EXPERIENCE

- **SUNHOU Architects & Partners Association** Taoyuan, Taiwan
Research & Development
– Lead digital transformation by developing internal RAG system, Revit plug-ins, Stable Diffusion and n8n framework Apr 2025 - Present
Working Student
– Executed a golf club interior renovation project (GFA 80,000 sqft) and an industrial factory project, including construction scheduling, detailed design, budget allocation, quality control, and MEP integration Sep 2020 - Jun 2022
- **ROCKS Interior Design** Taoyuan, Taiwan
Project Executive
– Led 2 major successes in architectural proposals, utilizing parametric design for complex geometric modeling, aiding in the re-planning of a 220,625 sq ft campus valued at US \$3.4M+ Jul 2022 - Aug 2022
- **OASISTUDIO Architect Office** Yilan, Taiwan
Intern
– Partnered with architects to secure 3+ projects, focused on the construction of historical building structures Jul 2020 - Aug 2020

LEADERSHIP & PROJECT EXPERIENCE

- **Real-Time Building Digital Model Reconstruction Using a Robotic Agent** (CCBE Software Lab 25') 2025
– Developed a real-time pipeline on Go2 robot (ROS2 Humble) for SLAM-based point cloud acquisition and processing
– Converted raw point cloud data into IFC-compliant semantic models automatically
- **Tree Species Classification & Multimodel Comparison** (Data Science in Earth Observation 25') 2025
– Benchmarked Random Forest, XGBoost, CNN, RNN, Transformer on Sentinel-2 data, evaluating predictive performance
- **Indoor Ambient Monitoring** (Geo Sensor Networks and the IoTs 25') 2025
– Integrated LoRaWAN time-series data via FROST API into real-time Cesium dashboard for comfort visualization
- **Interactive Visualization on Geothermal Energy** (Interactive Visualization 25') 2025
– Developed a visualized dome installation on geothermal energy and showcased at the 1E9 Festival der Zukunft in Munich
- **Constraint Design in Timber Modular Process** (AEC Hackathon Munich 24') 2024
– Developed a communication prototype for timber modular structures using PyRevit, collaborating with Drees & Sommer
- **Director, CYCU Architecture Digital Research Community** 2020–2023
– Organized 10+ lectures and workshops on digital fabrication, CAD/CAM and programming, benefiting 200+ participants
– Managed the digital fabrication laboratory for 2+ years, responsible for 50+ maintenance and operation tasks

AWARDS

- **Mark Winner, Golden Pin Design Award** 2023
- **Thesis Project Excellence Award**, Thermal Landscapes & Material Research 2023

SKILLS

- **Programming:** .Net Core (C#), HTML, CSS, JS, SQL, Python, Pandas, PyTorch | **Software:** Revit & Dynamo, Rhino & GH
- **Language:** English – Academic proficiency , German – Basic, Mandarin – Native language